```
import pandas as pd
```

```
emp=pd.read_excel(r'E:\FSDS\Dataset\Rawdata.xlsx')
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience#$ | 34 years | Mumbai | 5^00#0 | 2+ |
| **1** | Teddy^ | Testing | 45' yr | Bangalore | 10%%000 | <3 |
| **2** | Uma#r | Dataanalyst^^# | NaN | NaN | 1$5%000 | 4> yrs |
| **3** | Jane | Ana^^lytics | NaN | Hyderbad | 2000^0 | NaN |
| **4** | Uttam* | Statistics | 67-yr | NaN | 30000- | 5+ year |
| **5** | Kim | NLP | 55yr | Delhi | 6000^$0 | 10+ |

```
emp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       4 non-null      object
 3   Location  4 non-null      object
 4   Salary    6 non-null      object
 5   Exp       5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
id(emp)
```

```
1573323128528
```

```
emp.columns
```

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
emp.shape
```

```
(6, 6)
```

```
emp.isnull()
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False |
| **1** | False | False | False | False | False | False |
| **2** | False | False | True | True | False | False |
| **3** | False | False | True | False | False | True |
| **4** | False | False | False | True | False | False |
| **5** | False | False | False | False | False | False |

```
emp.isnull().sum()
```

```
Name        0
Domain      0
Age         2
Location    2
Salary      0
Exp         1
dtype: int64
```

```
emp.isna()
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False |
| **1** | False | False | False | False | False | False |
| **2** | False | False | True | True | False | False |
| **3** | False | False | True | False | False | True |
| **4** | False | False | False | True | False | False |
| **5** | False | False | False | False | False | False |

Data Cleaning

```
emp['Name']
```

```
0      Mike
1    Teddy^
2     Uma#r
3      Jane
4    Uttam*
5       Kim
Name: Name, dtype: object
```

```
emp['Name'] = emp['Name'].str.replace(r'\W','',regex=True)
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience#$ | 34 years | Mumbai | 5^00#0 | 2+ |
| **1** | Teddy | Testing | 45' yr | Bangalore | 10%%000 | <3 |
| **2** | Umar | Dataanalyst^^# | NaN | NaN | 1$5%000 | 4> yrs |
| **3** | Jane | Ana^^lytics | NaN | Hyderbad | 2000^0 | NaN |
| **4** | Uttam | Statistics | 67-yr | NaN | 30000- | 5+ year |
| **5** | Kim | NLP | 55yr | Delhi | 6000^$0 | 10+ |

```
emp['Name'] = emp['Name'].str.replace(r'\W','',regex=False)
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience#$ | 34 years | Mumbai | 5^00#0 | 2+ |
| **1** | Teddy | Testing | 45' yr | Bangalore | 10%%000 | <3 |
| **2** | Umar | Dataanalyst^^# | NaN | NaN | 1$5%000 | 4> yrs |
| **3** | Jane | Ana^^lytics | NaN | Hyderbad | 2000^0 | NaN |
| **4** | Uttam | Statistics | 67-yr | NaN | 30000- | 5+ year |
| **5** | Kim | NLP | 55yr | Delhi | 6000^$0 | 10+ |

```
emp['Domain'] = emp['Domain'].str.replace(r'\W','',regex=True)
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 years | Mumbai | 5^00#0 | 2+ |
| **1** | Teddy | Testing | 45' yr | Bangalore | 10%%000 | <3 |
| **2** | Umar | Dataanalyst | NaN | NaN | 1$5%000 | 4> yrs |
| **3** | Jane | Analytics | NaN | Hyderbad | 2000^0 | NaN |
| **4** | Uttam | Statistics | 67-yr | NaN | 30000- | 5+ year |
| **5** | Kim | NLP | 55yr | Delhi | 6000^$0 | 10+ |

```
emp['Age']=emp['Age'].str.extract('(\\d+)')
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5^00#0 | 2+ |
| **1** | Teddy | Testing | 45 | Bangalore | 10%%000 | <3 |
| **2** | Umar | Dataanalyst | NaN | NaN | 1$5%000 | 4> yrs |
| **3** | Jane | Analytics | NaN | Hyderbad | 2000^0 | NaN |
| **4** | Uttam | Statistics | 67 | NaN | 30000- | 5+ year |
| **5** | Kim | NLP | 55 | Delhi | 6000^$0 | 10+ |

```
emp['Salary']=emp['Salary'].str.replace(r'\W','',regex=True)
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5000 | 2+ |
| **1** | Teddy | Testing | 45 | Bangalore | 10000 | <3 |
| **2** | Umar | Dataanalyst | NaN | NaN | 15000 | 4> yrs |
| **3** | Jane | Analytics | NaN | Hyderbad | 20000 | NaN |
| **4** | Uttam | Statistics | 67 | NaN | 30000 | 5+ year |
| **5** | Kim | NLP | 55 | Delhi | 60000 | 10+ |

```
emp['Exp']=emp['Exp'].str.extract('(\\d+)')
```

```
emp
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5000 | 2 |
| **1** | Teddy | Testing | 45 | Bangalore | 10000 | 3 |
| **2** | Umar | Dataanalyst | NaN | NaN | 15000 | 4 |
| **3** | Jane | Analytics | NaN | Hyderbad | 20000 | NaN |
| **4** | Uttam | Statistics | 67 | NaN | 30000 | 5 |
| **5** | Kim | NLP | 55 | Delhi | 60000 | 10 |

```
clean_data = emp.copy()
```

```
id(clean_data)
```

```
1573323804016
```

```
clean_data.isnull()
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False |
| **1** | False | False | False | False | False | False |
| **2** | False | False | True | True | False | False |
| **3** | False | False | True | False | False | True |
| **4** | False | False | False | True | False | False |
| **5** | False | False | False | False | False | False |

```
import numpy as np
```

```
clean_data['Age']=clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
```

```
clean_data['Exp']=clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
```

```
clean_data['Location']=clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

```
clean_data
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5000 | 2 |
| **1** | Teddy | Testing | 45 | Bangalore | 10000 | 3 |
| **2** | Umar | Dataanalyst | 50.25 | Bangalore | 15000 | 4 |
| **3** | Jane | Analytics | 50.25 | Hyderbad | 20000 | 4.8 |
| **4** | Uttam | Statistics | 67 | Bangalore | 30000 | 5 |
| **5** | Kim | NLP | 55 | Delhi | 60000 | 10 |

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype 
---  ------    --------------  ----- 
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       6 non-null      object
 3   Location  6 non-null      object
 4   Salary    6 non-null      object
 5   Exp       6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
clean_data['Name']=clean_data['Name'].astype('category')
```

```
clean_data['Domain']=clean_data['Domain'].astype('category')
```

```
clean_data['Age']=clean_data['Age'].astype(int)
```

```
clean_data['Salary']=clean_data['Salary'].astype(int)
```

```
clean_data['Exp']=clean_data['Exp'].astype(int)
```

```
clean_data['Location']=clean_data['Location'].astype('category')
```

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype 
```

```
---  ------    --------------  -----
 0   Name      6 non-null      category
 1   Domain    6 non-null      category
 2   Age       6 non-null      int64
 3   Location  6 non-null      category
 4   Salary    6 non-null      int64
 5   Exp       6 non-null      int64
dtypes: category(3), int64(3)
memory usage: 938.0 bytes
```

```
clean_data.to_csv('clean_data.csv')
```

```
import os
os.getcwd()
```

```
'D:\\MyNotebook\\Test'
```

```
clean_data
```

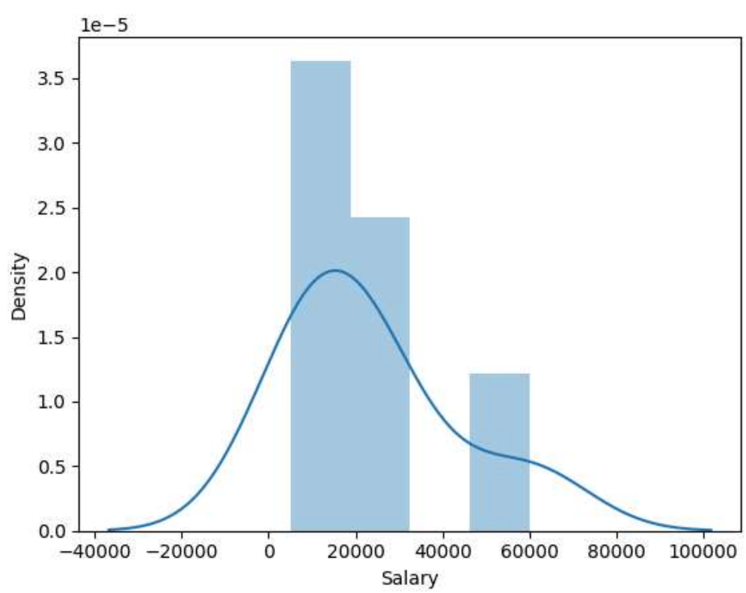
| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5000 | 2 |
| **1** | Teddy | Testing | 45 | Bangalore | 10000 | 3 |
| **2** | Umar | Dataanalyst | 50 | Bangalore | 15000 | 4 |
| **3** | Jane | Analytics | 50 | Hyderbad | 20000 | 4 |
| **4** | Uttam | Statistics | 67 | Bangalore | 30000 | 5 |
| **5** | Kim | NLP | 55 | Delhi | 60000 | 10 |

```
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

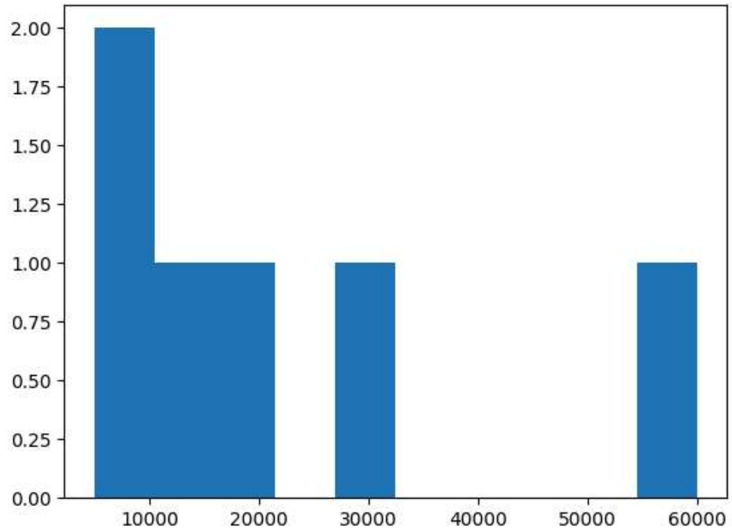
```
clean_data['Salary']
```

```
0     5000
1    10000
2    15000
3    20000
4    30000
5    60000
Name: Salary, dtype: int64
```

```
vis1 = sns.distplot(clean_data['Salary'])
```



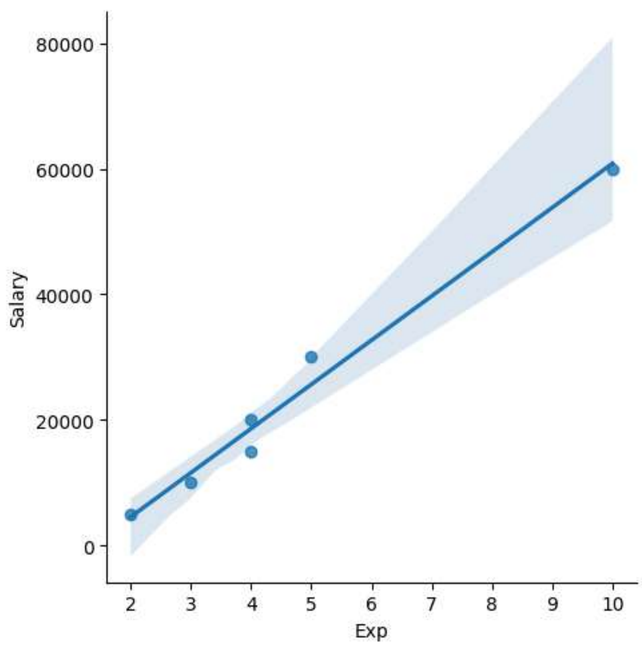
```
vis2 = plt.hist(clean_data['Salary'])
```



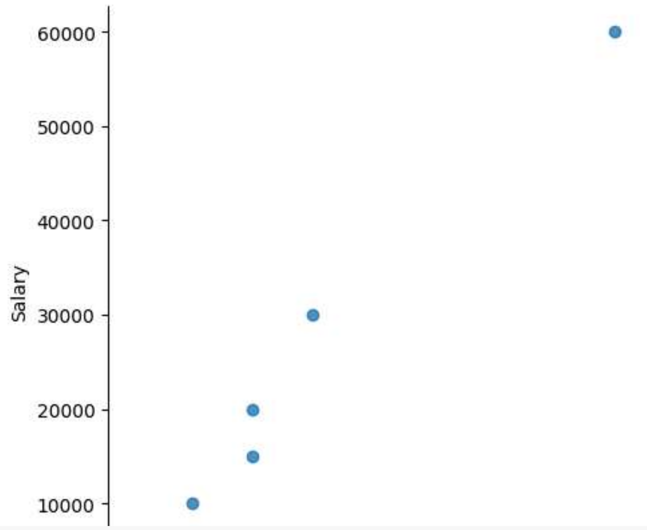

```
clean_data
```

| | Name | Domain | Age | Location | Salary | Exp |
|---|---|---|---|---|---|---|
| **0** | Mike | Datascience | 34 | Mumbai | 5000 | 2 |
| **1** | Teddy | Testing | 45 | Bangalore | 10000 | 3 |
| **2** | Umar | Dataanalyst | 50 | Bangalore | 15000 | 4 |
| **3** | Jane | Analytics | 50 | Hyderbad | 20000 | 4 |
| **4** | Uttam | Statistics | 67 | Bangalore | 30000 | 5 |
| **5** | Kim | NLP | 55 | Delhi | 60000 | 10 |

```
vis3 = sns.lmplot(data = clean_data,x ='Exp',y = 'Salary')
```




```
vis3 = sns.lmplot(data = clean_data,x ='Exp',y = 'Salary',fit_reg=False)
```




```
x_iv = clean_data[['Name','Domain','Age','Location','Exp']]
```

```
y_dv = clean_data[['Salary']]
```

```
imputation = pd.get_dummies(clean_data , dtype=int)
```

```
imputation
```

| | Age | Salary | Exp | Name_Jane | Name_Kim | Name_Mike | Name_Teddy | Name_Umar | Name_Uttam | Domain_Analytics | Domain_Dataanalyst | Domain_Data |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 34 | 5000 | 2 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | |
| **1** | 45 | 10000 | 3 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | |
| **2** | 50 | 15000 | 4 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | |
| **3** | 50 | 20000 | 4 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | |
| **4** | 67 | 30000 | 5 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | |
| **5** | 55 | 60000 | 10 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | |

from IPython.display import FileLink,FileLinks

```
Start coding or generate with AI.
```